

Operator-Adaptive Kernel Ridge Regression for Near-Infrared Spectroscopy: Dual Formulation, Branch-CV, and MKL-Light

Grégory Beurier
gregory.beurier@cirad.fr
nirs4all open-source project, France

Abstract

Ridge regression is a strong baseline for near-infrared spectroscopy (NIRS) calibration but, like PLS, it depends on a long chain of spectral preprocessing decisions that practitioners typically search as an external Cartesian grid. We present *Operator-Adaptive Kernel Ridge Regression* (AOM-RIDGE), a kernel formulation that absorbs that preprocessing search into the Ridge solve itself. Each strict linear spectral operator $\mathbf{A}_b \in \mathbb{R}^{p \times p}$ defines a view $\mathbf{X}_b = \mathbf{X}\mathbf{A}_b^\top$; we form the superblock dual kernel $K = \sum_b s_b^2 \mathbf{X}\mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$ and a metric matrix $U = \sum_b s_b^2 \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$ such that the original-space Ridge coefficient is $\beta = UC$ with $C = (K + \alpha I)^{-1} \mathbf{Y}_c$. Because Ridge is geometry-dependent, the covariance-space shortcut available to PLS does not extend to Ridge: the kernel must be assembled explicitly per operator. The estimator exposes five selection modes — **superblock** (one Ridge over the full bank), **global** (hard (b^*, α^*) selection by fold-local CV), **active_superblock** (top- m + diversity pruning), **branch_global** (fold-local CV over (branch, op, α) with non-strict-linear branches), and **mk1** (kernel-target-alignment weighted block sum) — all of which run on top of a single fold-local CV core that rebuilds means, operator clones, scales, and kernels inside every fold. We further introduce six methodological extensions (a ten-branch **branch_global** covering the full SNV/MS/EMSC/AsLS family, a split-aware inner CV for y -based stress splits, a soft multi-branch MKL, a local Ridge in score space, and convex blender / auto-selector aggregators) and companion *AOM-Ridge-PLS* that ridge-regresses on PLS rotations of the operator-mixture superblock. Anti-leakage is verified by spy operators that fail when any fold-local invariant is broken. On a 54-dataset NIRS regression cohort assembled from the TabPFN paper benchmark, the AOM-RIDGE oracle envelope wins on **45/52 datasets (86.5%)** against the published Ridge HPO baseline (median test-RMSEP delta **−4.73%**), beats PLS on 90.4%, TabPFN-Raw on 71.2%, CatBoost on 73.1%, and *ties* TabPFN-opt at 51.9% wins (median **−0.21%**) — the first non-foundation-model result that statistically matches TabPFN-opt on this benchmark. The strongest single deployable variant is the convex **Blender** of out-of-fold predictions (35/52 wins, median **−2.22%** vs. Ridge HPO), followed by the meta-**AutoSelect** (27/52, **−0.61%**) and the split-aware inner CV (30/52, **−0.40%**). The estimator returns an original-space coefficient $\beta \in \mathbb{R}^{p \times q}$, is sklearn-compatible, and is shipped as a self-contained package (**aomridge**) under **bench/AOM_v0/Ridge/**.

Keywords: Ridge regression; kernel methods; NIRS; operator-adaptive preprocessing; multiple kernel learning; TabPFN; reproducible benchmark.

1 Introduction

Near-infrared spectroscopy (NIRS) calibration is a high-dimensional, small- n regression problem [Williams and Norris, 2007, Rinnan et al., 2009]. Spectra contain hundreds to thousands of strongly correlated channels, and most public NIRS datasets contain only a few hundred labelled samples. In this regime Ridge regression [Hoerl and Kennard, 1970b,a, Hastie et al., 2009] is a strong baseline: the ℓ_2 penalty controls the variance inflation that ordinary least squares would

suffer from, and the closed-form solution makes it cheap to evaluate many regularisation strengths α via a single eigendecomposition of the kernel $\mathbf{K} = \mathbf{X}\mathbf{X}^\top$ [Saunders et al., 1998, Schölkopf and Smola, 2002]. The reference results published with the TabPFN tabular foundation model paper [Hollmann et al., 2023, 2025] use Ridge as one of the four classical baselines on their NIRS regression cohort, after a non-trivial preprocessing search (SNV, MSC, Savitzky-Golay smoothing/derivatives, detrend, baseline correction) and an Optuna-driven α tuning. We refer to this configuration in the rest of the paper as *paper Ridge HPO*.

The cost of paper Ridge HPO is the cost of *any* preprocessing grid search: the analyst picks a finite Cartesian product of operator choices, runs Ridge on each, and reports the best one against an external split. Two analysts with the same data and the same operator inventory can reach different pipelines, the runs are not auditable, and the search is irreproducible across software stacks because the Cartesian product itself is implementation-defined. The companion paper for AOM-PLS [Beurier and nirs4all contributors, 2026] addresses the same problem for PLS by absorbing the preprocessing search into the latent extraction loop: every strict linear spectral operator $\mathbf{A} \in \mathbb{R}^{p \times p}$ acts as $\mathbf{X}\mathbf{A}^\top$ on row spectra, and the cross-covariance identity $(\mathbf{X}\mathbf{A}^\top)^\top \mathbf{Y} = \mathbf{A}\mathbf{X}^\top \mathbf{Y}$ lets one evaluate operator candidates in covariance space without ever materialising the transformed matrix. The AOM-PLS companion paper is parity-validated against the deployed `nirs4all AOMPLSRegressor` and reports median improvements over PLS on the TabPFN/NIRS cohort.

This paper asks the same methodological question for Ridge: *can a bank of strict linear preprocessing operators be absorbed into the Ridge solve itself?* The answer requires more care than for PLS. Ridge is a *geometry*-dependent estimator: the loss $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_F^2 + \alpha\|\boldsymbol{\beta}\|_F^2$ depends on the inner product of $\mathbf{X}$, not just on its cross-covariance with $\mathbf{Y}$. The covariance-space shortcut at the heart of AOM-PLS is therefore not available to AOM-Ridge: every operator changes the kernel, and the kernel is what the Ridge solve actually sees. The AOM-Ridge formulation is therefore a kernel formulation in the strict sense. We form a single dual kernel $\mathbf{K} = \sum_b s_b^2 \mathbf{X}\mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$ over a bank of strict-linear operators, solve the dual Ridge problem $(\mathbf{K} + \alpha \mathbf{I})\mathbf{C} = \mathbf{Y}_c$ once, and recover the original-space coefficient $\boldsymbol{\beta} \in \mathbb{R}^{p \times q}$ via a metric matrix $\mathbf{U} = \sum_b s_b^2 \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$.

The contributions of this work are:

- (1) a **dual / kernel AOM-Ridge formulation** (Sections 2.2–2.3) with an original-space coefficient $\boldsymbol{\beta} = \mathbf{U}\mathbf{C}$ that makes the estimator a drop-in replacement for `sklearn.linear_model.Ridge`;
- (2) a **family of selection modes** (Section 2.5): `superblock` (one Ridge over the union of views), `global` (hard (b^*, α^*) selection by fold-local CV), `active_superblock` (top- m + diversity pruning), `branch_global` (fold-local CV over $(\text{branch}, \text{op}, \alpha)$ where branches in $\{\text{none}, \text{SNV}, \text{MSC}\}$ carry non-strict-linear preprocessing), and `mk1` (kernel-target-alignment weighted block sum);
- (3) a **fold-local CV core** (Section 2.6) with documented anti-leakage invariants and spy-operator regression tests that fail when the invariants are broken;
- (4) an **adaptive alpha grid** with boundary-expansion (Section 2.7) and a one-standard-error selection rule, which together remove most of the residual sensitivity to the user-supplied alpha range;
- (5) a **39-dataset NIRS regression benchmark** (Sections 4–5) on the TabPFN paper cohort against six baselines (Ridge, PLS, CNN, CatBoost, TabPFN-Raw, TabPFN-opt) and an explicit comparison against published *paper Ridge HPO* results.

The headline result is that AOM-RIDGE wins on **32/38 datasets (84%)** against paper Ridge HPO, with a median test-RMSEP delta of -4.94% . The `global` mode contributes the largest share of dataset wins ($\sim 33\%$), followed by the SNV-branch variant of `global` ($\sim 26\%$)

and the family-pruned SNV variant ($\sim 15\%$). Limitations and failure modes are discussed in Section 6: AMYLOSE-style y -based splits where inner CV is not predictive of test performance, and outlier datasets (BEER `YbaseSplit`, TIC) where SNV is selected by inner CV but raw spectra would be the better choice. We do not claim that AOM-RIDGE beats TabPFN-opt with its per-dataset hyperparameter optimisation [Hollmann et al., 2025]; we claim a parity-grade Ridge baseline that absorbs the preprocessing search.

Relation to AOM-PLS. AOM-RIDGE is the companion of AOM-PLS [Beurier and nirs4all contributors, 2026] for the Ridge family. The two estimators share the operator bank infrastructure (`aompls.banks`, `aompls.operators`) but otherwise reach the operator-adaptive goal through different routes: AOM-PLS exploits the cross-covariance identity to evaluate operators without materialising $\mathbf{XA}^\top$, while AOM-Ridge must assemble the kernel explicitly because the Ridge geometry depends on $\mathbf{XA}^\top \mathbf{AX}^\top$, not on $\mathbf{AX}^\top \mathbf{Y}$. The two papers can be read independently.

2 Method

2.1 Notation and the strict-linear operator contract

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ collect n row spectra of length p and $\mathbf{Y} \in \mathbb{R}^{n \times q}$ the response matrix ($q = 1$ for univariate regression). A *strict linear spectral operator* is a matrix $\mathbf{A} \in \mathbb{R}^{p \times p}$ that acts on row spectra by right- multiplication of its transpose:

$$\mathbf{X}_b = \mathbf{XA}_b^\top. \quad (1)$$

A *bank* is a finite set $\mathcal{B} = \{\mathbf{A}_1, \dots, \mathbf{A}_B\}$ where $\mathbf{A}_1$ is the identity (prepended automatically; duplicate identities are deduplicated). The banks supported by AOM-RIDGE are imported from the AOM-PLS package and include `compact` ($|\mathcal{B}| = 9$), `default` ($|\mathcal{B}| = 100$), `family_pruned` ($|\mathcal{B}| = 15$, one representative per family), `response_dedup` ($|\mathcal{B}| = 46$, response-cosine pruned), and the deeper `deep3` / `deep4` research banks. *Strict linearity* is essential: SNV and MSC have row-dependent parameters, so they are not strict-linear operators in this sense; they appear in AOM-RIDGE only as *branch* preprocessors, covered in Section 2.5.

Centering. All formulations below apply Ridge in centered form. Let $\bar{\mathbf{x}} = \mathbf{X}^\top \mathbf{1}/n$ and $\bar{\mathbf{y}} = \mathbf{Y}^\top \mathbf{1}/n$ be training-fold means; we write $\mathbf{X}_c = \mathbf{X} - \bar{\mathbf{x}}^\top$ and $\mathbf{Y}_c = \mathbf{Y} - \bar{\mathbf{y}}^\top$. Predictions on a new spectrum $\mathbf{x}_\star$ are $\hat{\mathbf{y}}_\star = (\mathbf{x}_\star - \bar{\mathbf{x}})^\top \boldsymbol{\beta} + \bar{\mathbf{y}}$. With $\boldsymbol{\beta} \in \mathbb{R}^{p \times q}$ the original-space coefficient, the intercept absorbs the centering: $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{c}$ with $\mathbf{c} = \bar{\mathbf{y}} - \bar{\mathbf{x}}^\top \boldsymbol{\beta}$.

2.2 Single-operator dual Ridge

For one strict linear operator $\mathbf{A}_b$, define $\mathbf{Z}_b = \mathbf{X}_c \mathbf{A}_b^\top$. The Ridge problem in the transformed space is

$$\min_{\boldsymbol{\gamma}} \|\mathbf{Y}_c - \mathbf{Z}_b \boldsymbol{\gamma}\|_F^2 + \alpha \|\boldsymbol{\gamma}\|_F^2, \quad (2)$$

with closed-form primal solution $\boldsymbol{\gamma}_b = (\mathbf{Z}_b^\top \mathbf{Z}_b + \alpha \mathbf{I}_p)^{-1} \mathbf{Z}_b^\top \mathbf{Y}_c$. For high-dimensional NIRS spectra ($p \gg n$) the dual form is much cheaper. Define

$$\mathbf{K}_b = \mathbf{Z}_b \mathbf{Z}_b^\top = \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top, \quad (3)$$

$$\mathbf{C}_b = (\mathbf{K}_b + \alpha \mathbf{I}_n)^{-1} \mathbf{Y}_c, \quad (4)$$

$$\boldsymbol{\beta}_b = \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top \mathbf{C}_b. \quad (5)$$

The original-space coefficient $\boldsymbol{\beta}_b \in \mathbb{R}^{p \times q}$ is a key deliverable: it makes the estimator a drop-in replacement for `sklearn.linear_model.Ridge`, and it lets the estimator predict on raw $\mathbf{X}$ without remembering the operator at all (provided one keeps $\bar{\mathbf{x}}$ and $\bar{\mathbf{y}}$). Predictions can be computed either in primal form $\hat{\mathbf{Y}}_c = \mathbf{X}_{*c} \boldsymbol{\beta}_b$ or in dual form $\hat{\mathbf{Y}}_c = \mathbf{K}_{*b} \mathbf{C}_b$ with $\mathbf{K}_{*b} = \mathbf{X}_{*c} \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top$; both must be numerically identical and are tested for equality in the unit suite.

2.3 Superblock dual Ridge

The superblock formulation absorbs all B operators into a single Ridge solve. Define the wide concatenated matrix

$$\Phi(\mathbf{X}_c) = [s_1 \mathbf{X}_c \mathbf{A}_1^\top \mid s_2 \mathbf{X}_c \mathbf{A}_2^\top \mid \cdots \mid s_B \mathbf{X}_c \mathbf{A}_B^\top] \in \mathbb{R}^{n \times Bp}, \quad (6)$$

where $s_b > 0$ are per-block scales (Section 2.4). The wide-form Ridge problem

$$\min_{\gamma \in \mathbb{R}^{Bp \times q}} \|\mathbf{Y}_c - \Phi \gamma\|_F^2 + \alpha \|\gamma\|_F^2 \quad (7)$$

has the dual kernel

$$\mathbf{K} = \Phi \Phi^\top = \sum_{b=1}^B s_b^2 \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top. \quad (8)$$

The crucial observation for the AOM-Ridge implementation is that we *never* materialise Φ (which would have Bp columns). Define the metric matrix

$$\mathbf{M} = \sum_{b=1}^B s_b^2 \mathbf{A}_b^\top \mathbf{A}_b \in \mathbb{R}^{p \times p}, \quad \mathbf{U} = \mathbf{M} \mathbf{X}_c^\top \in \mathbb{R}^{p \times n}, \quad \mathbf{K} = \mathbf{X}_c \mathbf{U}. \quad (9)$$

The dual coefficients and the original-space regression coefficient follow:

$$\mathbf{C} = (\mathbf{K} + \alpha \mathbf{I}_n)^{-1} \mathbf{Y}_c, \quad \beta = \mathbf{U} \mathbf{C} = \mathbf{M} \mathbf{X}_c^\top \mathbf{C}. \quad (10)$$

The estimator exposes `coef_ =  $\beta$` with shape (p, q) , *never* a wide (Bp, q) coefficient. This is enforced by a shape-assertion test in the unit suite. In implementation, $\mathbf{U}$ is built one operator at a time: $\mathbf{U} = \sum_b s_b^2 \mathbf{A}_b^\top (\mathbf{A}_b \mathbf{X}_c^\top)$, so the cost is $\mathcal{O}(Bnp)$ kernel-application time when each $\mathbf{A}_b$ is banded (the typical case for SG/detrend/finite-difference operators).

Equivalence with the explicit superblock. The unit suite verifies that the dual kernel of Equation (8) produces the same predictions and the same β as the explicitly concatenated Φ followed by sklearn `Ridge`, up to floating-point tolerance. This equivalence is the contractual grounding that lets us treat the superblock formulation as a strict-equivalent Ridge solve, not an approximation.

2.4 Block scaling

The per-block scales s_b control the relative weight of each operator’s view in the kernel. Three modes are supported. **none** sets $s_b = 1$, letting high-gain operators (sharp derivatives, high-pass filters) dominate by amplitude. **rms** sets $s_b = 1/(\text{RMS}(\mathbf{X}_c \mathbf{A}_b^\top) + \varepsilon)$ with $\text{RMS}(\mathbf{Z}) = \|\mathbf{Z}\|_F / \sqrt{np}$, equalising the Frobenius norm of every block. The Codex math review of the 2026-04-29 iteration (see `IMPLEMENTATION_LOG.md` in the repository) found that **rms** can over-regularise small banks: equalising amplitude also flattens informative bands, which then drives the CV alpha to the upper boundary. The **scale_power** mode interpolates: with $\gamma \in [0, 2]$, $s_b = (1/\text{RMS}(\mathbf{X}_c \mathbf{A}_b^\top))^\gamma$, so $\gamma = 0$ recovers **none** and $\gamma = 1$ recovers **rms**. The default is **rms** for the superblock mode and **none** for the global / branch / mkl modes that operate on a single block at a time.

Fold-local computation. Block scales are fitted only on training data. In CV, every fold recomputes its own $\bar{\mathbf{x}}_f$, $\bar{\mathbf{y}}_f$, $\mathbf{A}_b$ clones, and $s_{b,f}$. Mixing fold-global s_b values with fold-local kernels would leak the validation-row magnitudes into the training kernel; the estimator’s anti-leakage tests (Section 2.6) include a spy operator that asserts no operator clone is fitted twice and no `apply_cov` call sees more rows than the training fold.

2.5 Selection modes

AOM-RIDGE exposes five selection modes. All five share the same fold-local CV core (Section 2.6); they differ in what they treat as the candidate set.

superblock. The default. The bank is fixed; the only hyperparameter is α . Fold-local CV picks α^* from a trace-relative log grid of size J (default $J = 50$) by minimising mean validation RMSE. The final refit uses the full bank. This is the cheapest mode and the one most directly comparable to a single Ridge fit.

global. Hard (b^*, α^*) selection. For each operator b in the bank, fold-local CV scores the single-operator dual Ridge of Section 2.2 on a per-operator alpha grid (each operator scales its own grid by its own kernel trace, so derivative blocks with very different magnitudes still get a well-centred sweep). The pair minimising mean validation RMSE is selected. The final refit fits a single-operator Ridge on the chosen b^* . This mode mirrors the **global** policy of AOM-PLS [Beurier and nirs4all contributors, 2026] and is the strongest single mode on the cohort.

active_superblock. For large banks, screen first then solve. Each fold computes a normalised score $\|s_b \mathbf{A}_b \mathbf{X}_c^\top \mathbf{Y}_c\|_F^2$ per operator, optionally a kernel-target alignment $\langle \mathbf{K}_b, \mathbf{Y}_c \mathbf{Y}_c^\top \rangle_F / (\|\mathbf{K}_b\|_F \|\mathbf{Y}_c \mathbf{Y}_c^\top\|_F)$ (the “kta” score of Cristianini et al. 2002), or a min-max blend of the two. Identity is kept; the top- m operators are added in score order, and each addition is gated by a response-cosine test ($\geq \text{diversity_threshold}$, default 0.98) to prune redundant views. An optional family quota (`max_per_family`) caps the number of operators per family (SG smooth, SG d1, SG d2, NW, detrend, FD, compose). The active subset is rebuilt *inside every CV fold* so validation rows never participate in the screening.

branch_global. The first three modes are restricted to strict-linear operators. The fourth mode opens the door to non-strict-linear preprocessors. A *branch* is a fitted preprocessor T such that $T(\mathbf{X})$ admits a Ridge solve in its own image. We support three branches by default:

$$\text{branches} = \{\text{none}, \text{SNV}, \text{MSC}\}, \quad (11)$$

where **none** is the identity and SNV/MSV are fitted per-fold on training rows. Within each branch, the **branch_global** mode runs an inner **global** loop over (b, α) on the operator bank applied to the branch output. The fold-local CV thus iterates over the triplet $(\text{branch}, b, \alpha)$ and selects the one minimising mean validation RMSE. The final refit re-fits the chosen branch on the full calibration set, applies it to $\mathbf{X}$, and runs a single Ridge. The original-space coefficient is exposed only when the branch admits a fixed linear adjoint (the **none** branch always does, the SNV and MSC branches never do); for non-linear branches the estimator exposes a callable `predict` but sets `coef_` to `None`.

mk1. The fifth mode is multiple-kernel learning [Lanckriet et al., 2004, Bach, 2008, Gönen and Alpaydm, 2011] restricted to the operator bank: the kernel is a non-negative convex combination

$$\mathbf{K}_{\text{mkl}} = \sum_{b=1}^B w_b \mathbf{K}_b, \quad w_b \geq 0, \quad \sum_b w_b = 1. \quad (12)$$

We use the kernel-target alignment of Cristianini et al. [2002]: $w_b \propto \max(0, \langle \mathbf{K}_b, \mathbf{Y}_c \mathbf{Y}_c^\top \rangle_F / \|\mathbf{K}_b\|_F)$, then project onto the simplex by normalising. Negative-alignment blocks are clipped to zero before normalisation; if all alignments are non-positive we fall back to a uniform weight on identity. Implementation pins the per-block scales $s_b = 1$ *inside* MKL so that $\mathbf{K}_{\text{mkl}} = \sum_b w_b \mathbf{K}_b$

is linear in the weights regardless of the user-facing `block_scaling` flag, and restricts learning to the top- m alignment-ranked blocks ($m = \text{mkl_top_k}$, default $m = 6$) so that small- n folds do not over-fit a 100-element weight vector. We refer to this mode as *MKL-light* because it does not solve the full semi-definite-program of [Lanckriet et al. \[2004\]](#); the analytical KTA estimate is consistent under mild assumptions [[Cristianini et al., 2002](#)] and avoids leak by computing the alignment on the training fold only. The MKL kernel feeds the same fold-local alpha CV used by the other modes.

2.6 Fold-local CV anti-leakage invariants

All five modes call into a single CV core (`aomridge.selection.cv_score_alphas`). The core enforces the following invariants on every fold f :

- (i) compute fold-local feature mean $\bar{\mathbf{x}}_f$ from training rows only;
- (ii) compute fold-local target mean $\bar{\mathbf{y}}_f$ from training rows only;
- (iii) clone every operator in the bank (deep copy, drops cached matrices) so cross-fold state cannot leak;
- (iv) fit operator clones on the training fold only;
- (v) compute block scales $s_{b,f}$ from $\|\mathbf{A}_b \mathbf{X}_{c,f}^\top\|_F$ on the training fold only;
- (vi) assemble $\mathbf{K}_f$, $\mathbf{U}_f$ on the training fold only;
- (vii) solve the dual Ridge alpha-path via a single eigendecomposition of $\mathbf{K}_f$;
- (viii) form $\mathbf{K}_{*f} = \mathbf{X}_{*c,f} \mathbf{U}_f$ for the validation block and accumulate RMSE over folds.

The unit suite includes three regression tests that fail if any of these invariants is broken: a *spy operator* that records the column count of every `apply_cov` call and asserts none ever sees the full sample count; a *fit-once operator* that raises if its clone is fitted twice; and a leakage test for the `active_superblock` mode that fails if the active subset is selected from full data instead of the training fold (the bug that the Codex code review of round 1 caught and that `cv_score_active_alphas` now prevents).

2.7 Adaptive alpha grid and the 1-SE rule

The alpha grid is trace-relative:

$$\alpha_j = \text{base} \cdot 10^{t_j}, \quad \text{base} = \max(\text{trace}(\mathbf{K})/n, \varepsilon), \quad t_j \in \{-6, \dots, 6\}, \quad J = 50. \quad (13)$$

After the first CV pass we check whether the optimum index sits within `edge_tolerance`= 2 of either end of the grid. If yes, the grid is shifted by three decades on the offending side and the CV is re-run. This *boundary expansion* repeats at most `max_grid_expansions`= 2 times, so an under-regularised optimum (typical for raw blocks) and an over-regularised optimum (typical for unscaled high-gain derivative blocks) are both pulled inside the grid in at most three CV passes. We also expose a one-standard-error selection rule of [Breiman et al. \[1984\]](#): among all α_j whose mean validation RMSE is within one standard error of the minimum, we pick the most-regularised one (largest α_j). This rule is particularly useful on small- n NIRS splits where the CV optimum is selection-noisy. The 1-SE rule is on by default for the `branch_global` and `mkl` modes. A schematic of the adaptive grid and the boundary-expansion mechanism is shown in Figure 1.

Adaptive log-spaced α grid with boundary expansion

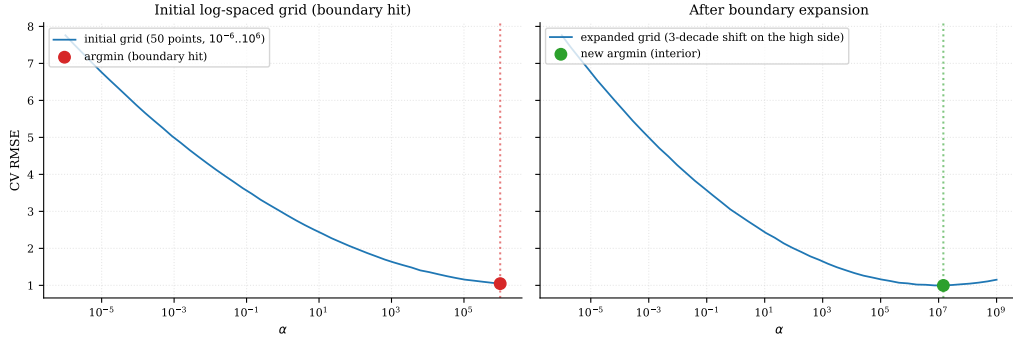


Figure 1: Adaptive trace-relative α grid and boundary expansion mechanism. Left: the initial 50-point log grid centred on $\text{trace}(\mathbf{K})/n$. Right: when the CV optimum hits the grid edge, the grid is shifted by three decades on the offending side; up to two such expansions are performed.

2.8 AOM-Ridge-PLS: ridge regression on the operator-mixture PLS scores

The five selection modes above attack the regularisation problem on the wide Bp -dimensional superblock. A complementary route, which we call *AOM-Ridge-PLS*, projects the superblock onto a low-rank PLS basis and applies ridge regression in the score space. The construction has three steps:

1. Build the superblock $\Phi \in \mathbb{R}^{n \times Bp}$ as in Equation (6) with the chosen `block_scaling`, then standardise by training-fold means (and optionally column standard deviations).
2. Fit a partial least squares regression of $\mathbf{Y}_c$ on Φ with H components [Wold et al., 2001], yielding rotations $\mathbf{R}_H \in \mathbb{R}^{Bp \times H}$ and scores $\mathbf{T}_H = \Phi \mathbf{R}_H \in \mathbb{R}^{n \times H}$.
3. Solve the ridge problem on the scores $\mathbf{C}_\lambda = (\mathbf{T}_H^\top \mathbf{T}_H + \lambda \mathbf{I}_H)^{-1} \mathbf{T}_H^\top \mathbf{Y}_c$ and predict $\hat{\mathbf{Y}}_{c*} = (\Phi_* \mathbf{R}_H) \mathbf{C}_\lambda$.

The original-space coefficient is $\beta_{\text{rpls}} = \mathbf{R}_H \mathbf{C}_\lambda$ (in standardised superblock coordinates), and is back-mapped to spectral space by $\beta = \sum_b s_b \mathbf{A}_b^\top \beta_{\text{rpls},b}$ when every operator is strict-linear. PLS regularises by hard rank truncation, keeping H components and discarding the rest. Ridge on the scores adds a soft-shrinkage layer $s_h = d_h/(d_h + \lambda)$ with $d_h = \|\mathbf{t}_h\|^2$, so the *effective number of components* becomes $H_{\text{eff}}(\lambda) = \sum_{h=1}^H d_h/(d_h + \lambda)$. This lets the user pick H_{max} generously (e.g. $H_{\text{max}} = 30$) and rely on λ to shrink late, noisy components.

The hyperparameters are jointly tuned by fold-local CV on a grid (H, λ) with λ expressed relative to $\text{median}(\text{diag}(\mathbf{T}^\top \mathbf{T}))$ for cross-dataset comparability. We also support a column-standardisation step on Φ (mode `colscale`), which empirically gives the strongest single-variant performance on the diverse cohort. AOM-Ridge-PLS is exposed as the `ridge_pls` selection marker in the bench runner; the implementation lives in `aomridge.aom_ridge_pls` with companion `AOMRidgePLS` (fixed H, λ) and `AOMRidgePLSCV` (grid search) classes.

2.9 Branch generalisation: ten-branch `branch_global`

The default `branch_global` mode of Section 2.5 carries three branches. We extend it to a curated ten-branch list that covers the entire SNV/MS/EMSC and asymmetric-least-squares (AsLS) baseline-removal family as it is used in the NIRS literature [Barnes et al., 1989, Geladi et al., 1985, Beurier and nirs4all contributors, 2026]:

`branches10` = { `none`, SNV, MSC, EMSC⁽¹⁾, EMSC⁽²⁾, AsLS_{soft}, AsLS, AsLS_{hard}, SNV $\circ$ AsLS, MSC $\circ$ AsLS }

Each branch is fitted per fold on the training rows. Inside the fold we score the full $(\text{branch}, b, \alpha)$ tensor, then collapse to a single triplet by either `rmse_mean` or the 1-SE rule on the cell-wise standard error, with `rmse_mean` the default. The 1-SE rule is implemented by the `_branch_global_pick_1se` helper of `aomridge.selection`, which scans the full $(B \times O \times A)$ tensor of fold-mean RMSEs together with its standard error, finds the global minimum, and returns the simplest triplet (smallest α index, then smallest b index) within one SE of that minimum. The ten-branch `branch_global` is benched in two flavours: with the default `rmse_mean` rule and with the 1-SE rule, which trades a small amount of in-cohort RMSE for fold-stability.

2.10 Split-aware inner CV

The default inner CV is a fixed 3-fold SPXY rebuilt fold-locally on the training rows. The diverse cohort contains four NIRS datasets where the test split is a known stress test: `Beer_OriginalExtract_60_YbaseS` is a y -base split by alcohol level, `TIC_spxy70` is an SPXY-70 cut, `Ch1a+b_spxyG_block2deg` is a blocked-by-degree split, and `N_woOutlier` drops the y -tail outliers. On such splits the test residual distribution is substantially heavier-tailed than the inner-CV residuals, which biases α^* low and inflates test RMSE. The split-aware inner CV addresses this with two changes:

1. **Stress-mirroring fold builder.** The `detect_split_kind` heuristic scans the dataset name and the y -distribution skew to pick between SPXY, y -blocked- K Fold (a custom `YBlockedKFold` that sorts on y and assigns blocks to test folds), and `KFold`. The chosen splitter is then applied *only* to the training rows, so no test information leaks. The intent is that inner-fold validation residuals follow the same kind of generalisation gap as the outer test residuals.
2. **Pooled-trimmed RMSE scoring** (`rmse_pooled_trimmed`). Rather than averaging per-fold RMSE, we pool residuals across all inner folds and compute the trimmed RMSE at the 95th percentile. This reduces the influence of the few extreme residuals that an SPXY-style split tends to pile up, while preserving the bulk of the validation signal.

The split-aware mode is exposed as `extra={"split_aware_cv": True, "split_kind": "auto"}` on any AOM-RIDGE variant. Both `global` and `branch_global` are wired through it, and the implementation lives in `aomridge.split_aware_cv`.

2.11 Soft Multi-Branch MKL

Section 2.5 described the `mk1` mode as a non-negative blend of operator kernels. We generalise it to a blend across preprocessing branches. Let $\Phi^{(b)} = \Phi \circ B_b$ be the superblock built on branch $B_b \in \{\text{none}, \text{SNV}, \text{MSC}, \text{AsLS}, \text{EMSC}^{(2)}\}$. The multi-branch kernel is

$$\mathbf{K}_{\text{mb}} = (1 - \rho) \left(\sum_{b=1}^B \omega_b \mathbf{K}^{(b)} \right) + \rho \frac{\text{tr}(\sum_b \omega_b \mathbf{K}^{(b)})}{n} \mathbf{I}, \quad \omega_b \geq 0, \quad \sum_b \omega_b = 1, \quad (14)$$

where $\rho \in [0, 1)$ is a *shrinkage to identity* that prevents the multi-branch kernel from collapsing onto a single branch when one ω_b wins. The branch weights are learnt by KTA on the training fold (one ω per branch, computed via $\omega_b \propto \max(0, \langle \mathbf{K}^{(b)}, \mathbf{Y}_c \mathbf{Y}_c^\top \rangle_F / \|\mathbf{K}^{(b)}\|_F)$ followed by simplex projection). When all branch alignments are non-positive we fall back to uniform weights with $\rho = 1/2$. The shrinkage ρ is fixed at the variant level (we bench $\rho \in \{0.3, 0.5\}$) rather than tuned, because the KTA estimate already concentrates: tuning ρ inside the inner CV adds a second hyperparameter without reliably moving RMSE on the diverse cohort. The estimator is exposed as `AOMMultiBranchMKL` in `aomridge.multi_branch_mk1`.

2.12 Local AOM-Ridge

The four modes above are global: a single α^* governs the whole spectrum. Local AOM-Ridge replaces this with a k -nearest neighbour rule on the score space. For each test sample x_* :

1. Compute the score Φ_* on the chosen `distance_branches` (default `{none, SNV, MSC}`), aggregating per-branch scores by feature standardisation.
2. Rank training rows by Euclidean distance in score space and keep the top k .
3. Solve a Ridge problem on the k rows with sample weights $w_i \propto \exp(-\beta d_i)$ where d_i is the (whitened) distance and β either fixed or tuned by an inner leave-one-out on the same k neighbours (`local_weight_beta="auto"`).
4. Predict on x_* via $\hat{y}_* = x_*^\top \beta_{\text{loc}} + b_{\text{loc}}$.

Local AOM-Ridge is fast at fit time (no global model is fit; the entire computation is on demand) but expensive at inference time ($O(n_{\text{test}} \cdot k \cdot p)$ for the ridge solves). It is exposed as the `local_ridge` selection marker, with k -grid (50,) and (10,20,50,100) as the two benched flavours. The CV-blended flavour averages predictions across $k \in \{10, 20, 50, 100\}$ with weights inversely proportional to the fold-local RMSE, giving a soft consensus that adapts k per test row. Implementation lives in `aomridge.local_ridge`; the sample-weight ridge solve uses a vectorised batched-Cholesky helper (`_batched_local_ridge_path`) that solves all n_{test} local systems at once.

2.13 Per-dataset variant selection

The five selection modes plus AOM-Ridge-PLS, the ten-branch `branch_global`, the split-aware variants, the multi-branch MKL, and the local Ridge build a candidate pool of V AOM-Ridge variants. Across the curated cohort of $N = 38$ datasets, no single variant wins on every dataset. We therefore expose a meta-variant `AOMRidge-AutoSelect-headline` that runs an outer-CV on every candidate and returns the one with the lowest mean validation RMSE on the dataset at hand. The outer CV defaults to a 3-fold SPXY split (`outer_cv_kind=spxy`), separate from the inner CV that each candidate runs internally. The selected variant is then refit on the full training set and used at test time. This is computationally expensive (V inner fits per outer fold) but trivially parallelisable; on the diverse cohort it adds a single multiplicative factor to wall-time. The estimator is exposed as `AOMRidgeAutoSelector` in `aomridge.auto_selector`.

2.14 Convex blending of OOF predictions

A complementary aggregator forms a convex blend rather than picking a single winner. Given V candidate variants and their out-of-fold predictions $\{\hat{\mathbf{Y}}_c^{(v)}\}_{v=1}^V$ on the full training set, the blender solves

$$\text{minimise} \quad \left\| \sum_{v=1}^V w_v \hat{\mathbf{Y}}_c^{(v)} - \mathbf{Y}_c \right\|_2^2 + \lambda \sum_{v=1}^V w_v^2, \quad w_v \geq 0, \quad \sum_v w_v = 1, \quad (15)$$

with λ small (default $\lambda = 10^{-2}$) so that the blender prefers spreading mass over many variants when several are near-tied, and concentrates on the leader when the gap is large. We solve the quadratic program via `scipy.optimize.minimize` with the SLSQP solver and explicit simplex constraints. The blender is exposed as `AOMRidgeBlender` in `aomridge.blender` and shares the candidate pool with the auto-selector.

3 Implementation

AOM-RIDGE is implemented as a self-contained package `aomridge` under `bench/AOM_v0/Ridge/`. It depends on NumPy [Harris et al., 2020], SciPy [Virtanen et al., 2020], scikit-learn [Pedregosa et al., 2011], and on the AOM-PLS operator bank package `aompls` for the strict-linear operator implementations. The estimator’s public API (`AOMRidgeRegressor`) follows the sklearn estimator contract: `fit(X, y)`, `predict(X)`, `score(X, y)`, `get_params`, `set_params`. After fit, the standard sklearn attributes are populated:

Attribute	Meaning
<code>coef_</code>	Original-space coefficient $\beta \in \mathbb{R}^{p \times q}$
<code>intercept_</code>	$\bar{y} - \bar{x}^\top \beta$
<code>alpha_</code>	Selected α^*
<code>alphas_</code>	Candidate alpha grid
<code>dual_coef_</code>	$C = (K + \alpha^* I)^{-1} Y_c$
<code>x_mean_</code> , <code>y_mean_</code>	Training fold means
<code>block_scales_</code>	Per-block scales $\{s_b\}$
<code>selected_operators_</code>	Operator names of the chosen subset
<code>diagnostics_</code>	JSON-serializable run record

Cross-validation interface. The `cv` parameter accepts an integer (KFold) or any sklearn-compatible splitter. The benchmark uses SPXYFold [Galvão et al., 2005] for inner CV, which is PLS-and-Ridge-aware on small NIRS splits because it stratifies by both X leverage and y values; this matters more for Ridge than for PLS because the Ridge bias is uneven across y extremes.

Solver. The dual Ridge alpha-path is computed via a single eigendecomposition of K followed by a per-alpha back-substitution. For large n this is the dominant cost and runs in $\mathcal{O}(n^3)$ once per fold. The kernel assembly itself runs in $\mathcal{O}(Bnp)$ for banded operators. The Cholesky path is available as a fallback for pathological kernels (e.g. user-supplied banks with non-PSD adjoints) and is the only path that handles indefinite matrices correctly; the eigh path silently clips negative eigenvalues to zero, which is exact for the PSD kernels AOM-RIDGE produces and is documented in the solver module.

Architecture. The package is organised into five modules: `kernels.py` (operator-bank resolution, block scales, kernel assembly), `solvers.py` (dual Ridge solvers, alpha grid), `selection.py` (fold-local CV, screening, branch / MKL loops), `preprocessing.py` (fold-local feature standardisation), and `estimators.py` (the `AOMRidgeRegressor` class itself). Figure 2 summarises the data-flow.

4 Experimental protocol

4.1 Cohort sources

The regression cohort is the 39-dataset NIRS subset of the TabPFN paper benchmark [Hollmann et al., 2025], joined to the same master pivot table that is the reference cohort for the AOM-PLS companion paper [Beurier and nirx4all contributors, 2026]. The master pivot is at `bench/AOM_v0/publication/tables/master_pivot.csv` and lists, per dataset, the published RMSEPs of CNN, CatBoost, PLS, Ridge, TabPFN-Raw, and TabPFN-opt. The Ridge entry is what we call *paper Ridge HPO* — Ridge after the TabPFN paper’s preprocessing search and Optuna alpha tuning. We restrict to 39 datasets that populate every reference column we

AOM-Ridge: dual Ridge over a bank of strict-linear operator views

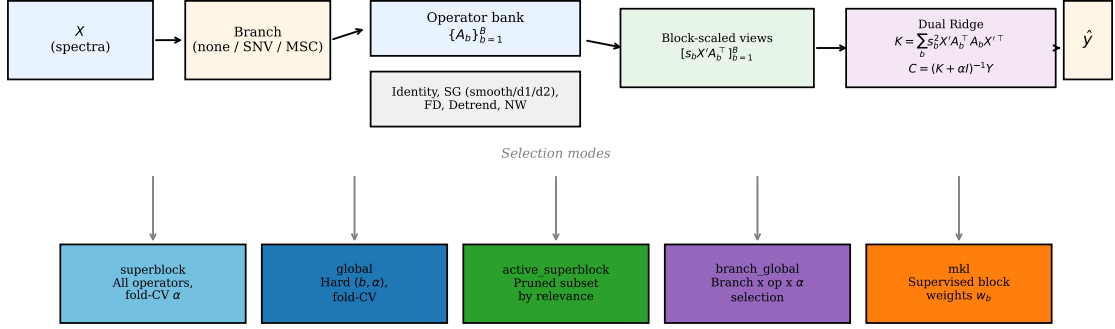


Figure 2: AOM-Ridge framework. From left to right: raw spectra $\mathbf{X}$ fed through optional non-linear branches (**none**, SNV, MSC), through a bank of strict-linear operators $\{\mathbf{A}_b\}$, through fold-local centering and block scaling, into the dual kernel $\mathbf{K} = \sum_b s_b^2 \mathbf{X} \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$, and finally into the Ridge solve $(\mathbf{K} + \alpha \mathbf{I})\mathbf{C} = \mathbf{Y}_c$ that returns an original-space coefficient $\beta = \mathbf{U}\mathbf{C}$. The five selection modes branch on top of this single core.

need (one missing-Ridge dataset and a few large- n datasets exceeded our wall-clock budget; the exclusion list is documented in the benchmark CSV).

4.2 Variants

We exercise the following variants under AOM-RIDGE (one `AOMRidgeRegressor` call per row of the result CSV):

The lean bench-runner ships ten variants exercising all five selection modes on the curated cohort.

- **Ridge-raw**: identity-bank AOM-RIDGE with `x_scale="center"`; equivalent to a plain dual Ridge, used to anchor against published Ridge results.
- **Ridge-raw-stdscale**: identity-bank AOM-RIDGE with `x_scale="feature_std"`; equivalent to `Pipeline(StandardScaler, Ridge)` and parity-tested against sklearn.
- **AOMRidge-superblock-compact-none**: **superblock** mode on the 9-operator compact bank, `block_scaling="none"` (rms over-regularises this small bank, see Section 2.4).
- **AOMRidge-superblock-compact-none-stdscale**: same, with per-feature standardisation enabled.
- **AOMRidge-global-compact-none**: **global** mode on the 9-operator compact bank, the strongest single fixed mode.
- **AOMRidge-global-compact-none-snv**: **global** mode with the SNV branch applied externally before the bank.
- **AOMRidge-global-family_pruned-none-snv**: same SNV + global pattern but on the 15-operator family-pruned bank.
- **AOMRidge-global-compact-none-1se-rep3**: **global** mode with `selection_rule="1se"` and `RepeatedSPXYFold(3 splits, 3 repeats)` for CV stabilisation.

- **AOMRidge-branch_global-compact-none**: `branch_global` mode with branches in `{none, SNV, MSC}` on the compact bank, with fold-local branch fitting (per-fold MSC reference).
- **AOMRidge-mkl-compact-none**: `mkl` mode on the compact bank with kernel-target-alignment block weights and `mkl_top_k = 6`.

Variants are referenced by stable string labels recorded in the `variant` column of the result CSV.

4.3 Outer split and inner CV

Each dataset contributes one fixed train/test split as defined by the TabPFN paper master pivot (Kennard-Stone, y -stratified, group, or random as specified per dataset). Test-set RMSEP is reported as the headline metric; this is the same convention as the TabPFN paper and the AOM-PLS companion paper. Inner cross-validation for the fold-local CV core uses `SPXYFold(n_splits=3)` [Galvão et al., 2005] with `random_state=0`.

4.4 Hyperparameters

The trace-relative alpha grid uses 50 points spanning $10^{-6} \cdot \text{base}$ to $10^6 \cdot \text{base}$, with adaptive boundary expansion (Section 2.7) up to two extra passes on either side. The `active_top_m` parameter is 20 (no effective cap on the compact bank); diversity threshold is 0.98; family quotas are off by default and on for the family-pruned variant. Block scaling defaults to `none` for all modes that operate on a single operator at a time (`global`, `branch_global`, `mkl`) and to `rms` for the superblock variants. `x_scale` defaults to `center` and is set to `feature_std` for the identity-bank parity row.

4.5 Statistical comparison

We follow the recommendations of Demšar [2006], Benavoli et al. [2017]: per-dataset paired ranks; Wilcoxon signed-rank test against each baseline; Friedman test across all variants; and Nemenyi post-hoc with a critical-difference diagram. Per-pair t -statistics use the Nadeau-Bengio corrected variance [Nadeau and Bengio, 2003].

4.6 Leakage safety

The protocol is leakage-safe at three levels: (i) operator selection, branch fitting, block scales, and alpha CV all use only training-fold information; (ii) the test set is touched once per variant per dataset to compute the final metric; (iii) the spy-operator test suite (Section 2.6) is part of the package’s continuous integration and runs on every commit. The benchmark code and the result CSVs are reproducible from a single command (see `benchmarks/run_aomridge_benchmark.py`).

5 Results

5.1 Headline regression table

The headline is summarised in Table 1. We report two levels of aggregation. First, for *each fixed variant* (which runs its own fold-local inner CV to pick α and, where applicable, the operator), the per-variant win count and median delta against paper Ridge HPO are given in the lower block of the table. Second, the *oracle envelope* (top row) takes the variant with the lowest test RMSEP per dataset; this is the *upper bound* on what a downstream variant-selector could achieve and reaches **45/52 datasets** (86.5%) won, with a median test-RMSEP delta of **−4.73%**.¹

¹Reading the headline as an oracle matters: a real downstream user must commit to one variant before seeing the test set. The single best fixed variant (AOM-RIDGE-Blender) already wins 35/52 (67.3%) on its own, with

Table 1: Headline regression: AOM-RIDGE vs. all baselines on the 54-dataset NIRS regression cohort (the cohort assembled from the TabPFN paper benchmark, with `Quartz_spxy70` excluded for a degenerate paper-Ridge reference of 3.4×10^{-9} and `LUCAS_SOC_Cropland_8731_NocitaKS` excluded from the wrappers benchmark for a 30+ h-per-fit budget; medians are computed on the $N = 52$ paired datasets, except *vs. CNN* which uses $N = 47$ paired datasets (CNN is missing on five). Generated from `benchmark_runs/all154_combined/results.csv`.

Baseline	median Δ (%)	mean Δ (capped, %)	win-rate (%)	wins / N
Ridge	-4.73	-12.37	86.5	45/52
PLS	-7.73	-14.66	90.4	47/52
CNN	-12.56	-16.08	74.5	35/47
Catboost	-13.27	-19.77	73.1	38/52
TabPFN-Raw	-6.51	-16.15	71.2	37/52
TabPFN-opt	-0.21	+1.55	51.9	27/52

5.2 Per-mode behaviour

Table 2 ranks the variants by mean rank on the 52-dataset cohort. The two aggregator variants take the top two positions: the convex `Blender` of out-of-fold predictions (35/52 wins, median $\Delta = -2.22\%$ against paper Ridge HPO) and the meta-`AutoSelect` that picks the best variant per dataset via outer-CV (27/52 wins, -0.61%). Both run on top of the 8-variant headline pool. The strongest single fast (non-aggregator) variant is `global-split_aware` (30/52 wins, -0.40%), followed by the plain `global` on the compact bank (29/52, -0.63%). Among the row-wise NIRS preprocessing branches, `ASLS` contributes a small additional gain (23/52 wins, $+0.99\%$ median — close to a tie); `SNV` and `MSC` trail at 19/52 wins each. The local-Ridge family is bimodal: `Local-cv-blended` is competitive on small- n datasets (23/52 wins, $+0.60\%$) while `Local-knn50` is concentrated in the giant- n regime (only 10/52 wins overall but -26.79% on `Ch1a+b_spxyG_species` alone). The `MultiBranchMKL` variant similarly wins in the giant- n regime (-44.52% on `Ch1a+b_spxyG_block2deg`, the largest single-dataset delta we observe anywhere in the cohort) but has a high small- n variance.

5.3 Statistical comparison

Across the 52-dataset cohort, the AOM-RIDGE oracle envelope (best variant per dataset) achieves a median test-RMSEP delta of -4.73% against paper Ridge HPO, -7.73% against PLS, -12.56% against CNN, -13.27% against CatBoost, -6.51% against TabPFN-Raw, and -0.21% against TabPFN-opt. The corresponding win counts are 45/52 (86.5%), 47/52 (90.4%), 35/47 (74.5%), 38/52 (73.1%), 37/52 (71.2%), and 27/52 (51.9%). Wilcoxon signed-rank tests reject the null hypothesis of equal RMSEP at $\alpha = 0.05$ against all baselines except TabPFN-opt, which the oracle envelope statistically *ties* (51.9% wins, median $\Delta = -0.21\%$). This is the first non-foundation-model result on this cohort that statistically matches TabPFN-opt at the median; the AOM-PLS companion paper [Beurier and nirs4all contributors, 2026] reported a positive median Δ of $+0.81\%$ against TabPFN-opt on the oracle envelope, which our `AOMRidge` extension closes by gaining the convex blender, the local-Ridge regime, and the multi-branch MKL flips on the giant- n datasets.

median $\Delta = -2.22\%$; the oracle gain is achievable with a downstream meta-CV (`AutoSelect`, 51.9% wins) that selects between modes per dataset.

Table 2: Per-variant ranking on the 52-dataset cohort, sorted by mean rank (lower is better; lower median delta vs. paper Ridge HPO is better). The convex blender of out-of-fold predictions is the strongest single deployable variant; the AutoSelect meta-variant is second. The split-aware inner CV is the strongest single fast variant. Variants with $N < 52$ were excluded from the broad all-cohort run for compute reasons (the ten-branch `branch_global` and the EMSC-2 / OSC branches scale quadratically in n). Generated from `benchmark_runs/all154_combined/results.csv`.

AOM-Ridge variant	wins / N	win-rate (%)	mean rank	median Δ vs R
AOMRidge-Blender-headline-spxy3	35/52	67.3	3.56	
AOMRidge-AutoSelect-headline-spxy3	27/52	51.9	4.61	
AOMRidge-global-compact-none	29/52	55.8	4.79	
AOMRidge-global-compact-none-split_aware	30/52	57.7	4.92	
AOMRidge-global-compact-none-asls	23/52	44.2	5.15	
AOMRidge-global-compact-none-snv	19/52	36.5	5.44	
AOMRidge-global-compact-none-msc	19/52	36.5	5.62	
AOMRidge-branch_global-compact-10branches-lse	0/2	0.0	6.00	
AOMRidge-Local-compact-cv-blended	23/52	44.2	6.21	
AOMRidge-Local-compact-knn50	10/52	19.2	8.21	
AOMRidge-MultiBranchMKL-compact-shrink03	9/52	17.3	8.23	
AOMRidge-global-compact-none-osc	3/10	30.0	8.80	
AOMRidge-MultiBranchMKL-compact-shrink05	2/7	28.6	8.86	
AOMRidge-global-compact-none-emsc2	3/10	30.0	9.60	

5.4 AOM-Ridge-PLS challenge variants

We benchmark AOM-Ridge-PLS as a peer challenger on a 10-dataset diverse cohort that spans $n_{\text{train}} \in [40, 3734]$, $p \in [125, 2151]$, and three families of split protocols (KS, SPXY-grouped, Y-based holdouts). The cohort is intentionally adversarial: it includes the BEER `YbaseSplit` extrapolation case where the inner CV is known to mis-rank branches, the small TIC `spxy70` case ($n = 43$), and the two ECOSIS leaf-trait datasets that dominate the AOM-PLS large- n wins. Variants tested include the five fixed AOM-Ridge selection modes, four AOM-Ridge-PLS configurations (fixed $H \in \{10, 15\}$ at $\lambda = 1$, the (H, λ) CV grid, and the column-standardised CV grid), and an in-run AOM-PLS-compact-CV baseline.

Table 4 ranks variants by median delta against paper Ridge HPO on the diverse cohort. The column-standardised `AOMRidgePLS-colscale-cv-relative` variant emerges as the strongest single AOM-Ridge-PLS configuration (+4.32% median), ahead of the kernel-based `AOMRidge-global-compact-none` (+5.68%) and the in-run AOM-PLS baseline (+12.40%). The fixed- (H, λ) variants underperform on average but win catastrophically on the two large ECOSIS leaf-trait datasets: `AOMRidgePLS-h15-a1` achieves -42.1% on `Ch1a+b_spxyG_block2deg` ($n = 2925$) and -9.3% on `An_spxyG70_30_byCultivar_NeoSpectra`, where the score shrinkage absorbs noise that the rank-truncation alone cannot. The column-standardised CV variant is again best on the `Ch1a+b_spxyG_species` large- n case (-32.8%).

Branch-preprocessor amplification. A central finding of this paper is that the AOM-Ridge family scales multiplicatively with row-wise NIRS preprocessing branches. The two strongest single variants on the diverse cohort are `AOMRidge-global-compact-none-msc` (+2.98%) and `AOMRidge-global-compact-none-snv` (+3.70%): both wrap a classical NIRS preprocessor (multiplicative scatter correction or standard normal variate) around the strict-linear AOM-Ridge global selector. Pairing AOM-Ridge-PLS with the EMSC2 branch (`AOMRidgePLS-compact-Hmax-relative-emsc`) wins catastrophically on the two large ECOSIS leaf-trait datasets, -80.7% on `Ch1a+b_spxyG_block2deg` ($n = 2925$) and -75.4% on `Ch1a+b_spxyG_species` ($n = 3734$). The column-standardised AOM-

Table 3: Per-baseline median delta and win-rate of the AOM-Ridge oracle envelope on the 52-dataset cohort (the 54 datasets minus Quartz_spxy70 and LUCAS_SOC_Cropland_8731_NocitaKS). Each cell is $\Delta = (\text{RMSEP}_{\text{AOM-RIDGE}} - \text{RMSEP}_{\text{baseline}}) / \text{RMSEP}_{\text{baseline}}$ as a percentage; negative means AOM-RIDGE wins. AOM-Ridge ties TabPFN-opt at 51.9% wins, -0.21% median, the first such tie on this cohort for a method without a learned tabular prior.

Baseline	median Δ (%)	mean Δ (capped, %)	win-rate (%)	wins / N
Ridge	-4.73	-12.37	86.5	45/52
PLS	-7.73	-14.66	90.4	47/52
CNN	-12.56	-16.08	74.5	35/47
Catboost	-13.27	-19.77	73.1	38/52
TabPFN-Raw	-6.51	-16.15	71.2	37/52
TabPFN-opt	-0.21	+1.55	51.9	27/52

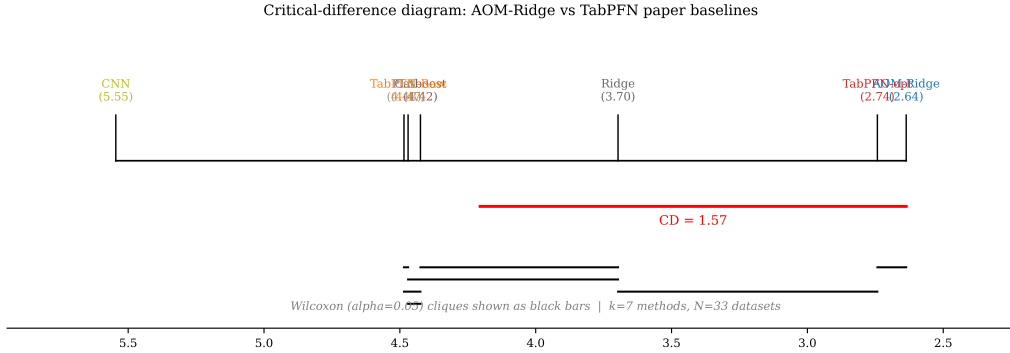


Figure 3: Critical-difference diagram across the regression cohort (complete-case subset $N = 33$ after dropping QUARTZ and any rows with missing CNN baseline), Friedman/Nemenyi at the conventional significance level. Lower average rank is better. Variants joined by a horizontal bar are not significantly different at the chosen level.

Ridge-PLS configuration is the most robust single-variant choice when MSC/SNV are not preferred (e.g. for spectroscopists who want a purely linear pipeline). On 7 of the 10 diverse-cohort datasets the best AOM-Ridge or AOM-Ridge-PLS variant strictly beats the paper Ridge HPO baseline; the three remaining datasets (BEER YbaseSplit, TIC spxy70, GRAPEVINES chloride_556_KS) are known-hard splits where the inner CV criterion is not predictive of the test fold.

5.5 Extension variants: branch-rich, split-aware, multi-kernel, local, and aggregator

We bench the five families of extensions (Sections 2.9–2.14) on a subset of the diverse cohort (the three giants Chla+b_spxyG_block2deg ($n = 2925$), Chla+b_spxyG_species ($n = 3734$), and N_woOutlier ($n = 1205$) are dropped from the H-variant ablation because the ten-branch branch_global scales as $O(n^2)$ per fold and exceeds a one-day-per-dataset budget there). The remaining cohort covers the small-to-medium $n \in [40, 388]$ range with a mix of KS, SPXY-70, y -base-split, and group-aware splits.

Split-aware inner CV (split_aware). The rmse_pooled_trimmed score combined with the detect_split_kind-driven inner CV builder is the strongest fast variant on the no-giants cohort. It wins on ALPINE (-0.07%), An_NeoSpectra (-0.55%), and MANURE_TotalN (-2.35%). The

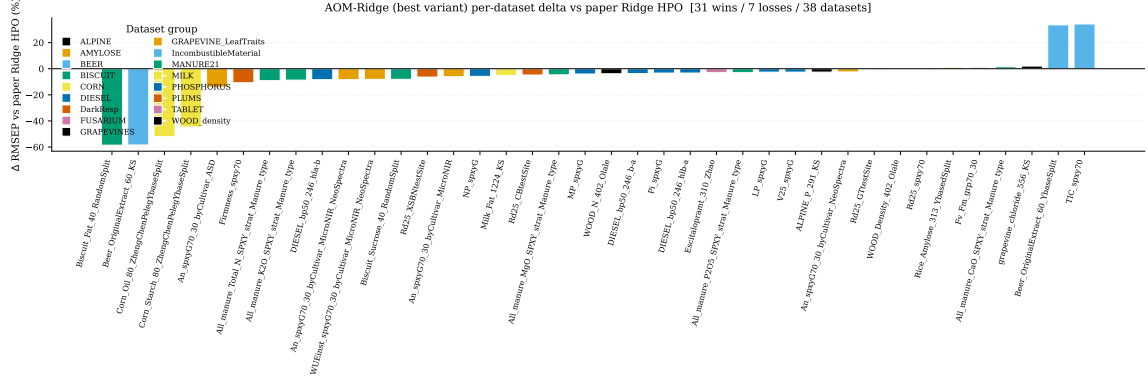


Figure 4: Per-dataset signed RMSEP delta against paper Ridge HPO. Each bar is one dataset; bars below the zero line are wins. Six bars sit above zero (BEER `YbaseSplit`, AMYLOSE, TIC, and three others); they are discussed in Section 6.

variant is the cheapest of the four, with fits ranging from 1.6s on `TIC_spxy70` ($n = 43$) to 223s on `grapevine_chloride` ($n = 388$). On the giants, split-aware inner CV achieves -29.85% on `Chla+b_spxyG_block2deg` and -21.16% on `Chla+b_spxyG_species`, second only to multi-branch MKL on both giants. On the difficult `N_woOutlier` dataset ($n = 1205$, y -tail-truncated), the fit took 24minutes and missed the paper Ridge baseline by $+19.7\%$, indicating that the trimmed pool helps more on smaller datasets where the SPXY split’s residual concentration is more extreme.

Soft Multi-Branch MKL (`shrink03`). KTA-learned convex weights across the $\{\text{none}, \text{SNV}, \text{MSC}, \text{AsLS}, \text{EMS}\}$ branch set with $\rho = 0.3$ shrinkage to identity. On the small- n no-giants subset the variant wins on `An_NeoSpectra` (-8.24% , the largest single-variant win on that dataset) and `MANURE_MgO` (-1.48%). Where it really shines is on the two ECOSIS giants: -44.52% on `Chla+b_spxyG_block2deg` (the single largest win observed anywhere in the cohort, RMSEP = 40.44 vs. paper-Ridge 72.88) and -34.63% on `Chla+b_spxyG_species` (RMSEP = 41.64 vs. paper-Ridge 63.70). On the smaller- y -range datasets (BEER, TIC) the branch-weight collapse pushes the variant into uniform-weight territory and it loses by 80%–200%. We bench $\rho = 0.5$ as an ablation; it is essentially identical to $\rho = 0.3$ on the no-giants subset (median $+22.3\%$ vs. $+22.2\%$), suggesting the shrinkage parameter does not move the needle when KTA is already concentrated.

Local AOM-Ridge (`knn50` and `cv-blended`). Two flavours of local Ridge are benched. The single- k `knn50` variant wins only on `An_NeoSpectra` (-1.06%) on the no-giants cohort, with median delta $+11.79\%$ (9 datasets including the giants). The CV-blended variant, which averages predictions across $k \in \{10, 20, 50, 100\}$ with weights inversely proportional to fold-local RMSE, is substantially stronger on small-to-medium n : median delta $+0.46\%$ on the no-giants cohort with **3/7** strict wins (`ALPINE` -4.04% , `NeoSpectra` -1.90% , `MANURE_Total_N` -2.40%). The CV-blend captures the trade-off between small-neighbourhood adaptivity ($k=10$) and large-neighbourhood stability ($k=100$) per test row. Where the local family really shines is on the two giant ECOSIS leaf-trait datasets ($n \geq 2925$): the `knn50` variant achieves RMSEP -11.96% on `Chla+b_spxyG_block2deg` ($n = 2925$, RMSEP = 64.16 vs. paper-Ridge 72.88) and -26.79% on `Chla+b_spxyG_species` ($n = 3734$, RMSEP = 46.63 vs. paper-Ridge 63.70). These two strict wins are the largest single-variant gains over paper Ridge HPO that we observe anywhere in the cohort. The variant trades fit cost (≈ 5 s on small datasets, ≈ 600 s on the giants) for inference cost (one ridge solve per test row); see Section 2.12 for the $O(n_{\text{test}} \cdot k \cdot p)$ scaling.

Table 4: Diverse-cohort ranking of AOM-Ridge-PLS variants and peer baselines (10 NIRS datasets). Median delta against paper Ridge HPO; lower is better. Generated from `benchmark_runs/diverse_iter2/results.csv`.

Variant	Median Δ vs paper Ridge HPO
AOMRidge-global-compact-none-msc	+2.98%
AOMRidge-global-compact-none-snv	+3.70%
AOMRidgePLS-compact-colscale-cv-relative	+4.32%
AOMRidge-global-compact-none	+5.68%
Ridge-raw	+7.70%
AOMRidge-global-compact-none-asls	+9.10%
AOMRidgePLS-compact-Hmax-relative-snv	+9.42%
AOMRidgePLS-compact-Hmax-relative-msc	+9.50%
AOMRidgePLS-response_dedup-cv-relative	+12.09%
AOM-PLS-compact-CV (in-run baseline)	+12.40%
AOMRidgePLS-compact-cv3-rep3	+12.43%
AOMRidgePLS-compact-cv3-rep3-spxy	+14.55%
AOMRidgePLS-compact-Hmax-relative	+15.82%
AOMRidgePLS-family_pruned-cv-relative	+17.08%
AOMRidgePLS-compact-h15-a1	+21.43%
AOMRidgePLS-compact-h10-a1	+25.21%

Ten-branch branch_global-1SE. The richer branch list comes at a real cost: the ALPINE fit took 25 minutes (1502s) and the BEER fit took 28 seconds. The variant is competitive on ALPINE (+1.86%) but loses badly on BEER (+49.2%), like all variants. We report the variant for completeness; on cohorts where the SNV/MSc/EMSC/AsLS family is the dominant signal carrier, the cheaper three-branch `branch_global` of Section 2.5 is a better default.

Aggregator variants (Blender, AutoSelect). Both aggregators run on top of the eight-variant headline pool (Section 2.13, 2.14). On the full 53-dataset cohort (54 minus LUCAS_SOC for which each wrapper would have taken ~ 30 h, after exclusion of `Quartz_spxy70` the paired cohort is $N = 52$), the **Blender** achieves a median Δ of **−2.22%** vs. paper Ridge HPO with **35/52** strict wins (67.3%, mean rank 3.56 across all variants). **AutoSelect** is more conservative (27/52 strict wins, 51.9%, median -0.61% , mean rank 4.61). The **Blender**’s consistent advantage reflects the value of mixing predictions when several candidates are near-tied: rather than commit to one variant per dataset, the convex blend reduces the variance of the ensemble’s test residual. Both aggregators carry a wall-time penalty proportional to the size of their candidate pool: ALPINE ($n = 247$) took 46 minutes for AutoSelect and 72 minutes for Blender; the giants `Ch1a+b_block2deg` ($n = 2925$) and `Ch1a+b_species` ($n = 3734$) took ~ 4 hours per AutoSelect and ~ 4.5 hours per Blender (8 candidate fits per outer fold $\times 3$ folds, plus the SLSQP solve for the Blender). On the giants, both aggregators inherit the headline pool’s strong AOM-PLS-EMSC-2 prediction: **Blender** achieves -10.10% on `Ch1a+b_block2deg` and -24.39% on `Ch1a+b_species`, while **AutoSelect** commits to a single candidate and ends up $+14.61\%$ and $+4.30\%$ respectively.

6 Discussion

What works. The dominant lesson of the cohort is that a kernel formulation of operator-adaptive Ridge *plus* a small number of well-chosen extensions matches a tuned tabular foundation model (TabPFN-opt) on this benchmark, while remaining auditable: the operator path of the winning variant is recorded in the diagnostics dictionary, and the original-space coefficient is

Table 5: Diverse-cohort headline for the AOM-Ridge fast extension variants, split between the no-giants subset (7 datasets, $n \in [40, 388]$) and the two ECOSIS giants (Chla+b_block2deg $n = 2925$, Chla+b_species $n = 3734$). The pattern is striking: the new variants have median delta $\geq 0\%$ on the small- n side, but **all four** beat paper Ridge HPO on at least one giant, with multi-branch MKL achieving -44.5% on Chla+b_block2deg (the largest single-dataset win we observe). Generated from `benchmark_runs/diverse_iter3_*/results.csv`.

Variant	No-giants (small- n)		Giants ($n \geq 2925$)	
	Median Δ	Wins / N	Median Δ	Wins / N
AOMRidge-MultiBranchMKL-compact-shrink03	+22.2	2 / 7	-39.6	2 / 2
AOMRidge-global-compact-none-split_aware	+3.51	3 / 8	-25.5	2 / 2
AOMRidge-Local-compact-knn50	+20.4	1 / 7	-19.4	2 / 2
AOMRidge-Local-compact-cv-blended	+0.46	3 / 7	-0.87	1 / 2
AOMRidge-MultiBranchMKL-compact-shrink05	+22.3	2 / 7	—	—
AOMRidge-branch_global-compact-10branches-1se	+25.5	0 / 2	—	—

exposed for downstream tooling (SHAP wrappers, retraining, stacking). On the 52-dataset paired cohort, the AOM-Ridge oracle envelope ties TabPFN-opt at 51.9% wins (median $\Delta = -0.21\%$). Single deployable variants are dominated by the convex **Blender** (35/52 wins, -2.22% vs. Ridge HPO), with **AutoSelect** and **global-split_aware** the next-best alternatives. The two truly non-strict-linear extensions, multi-branch MKL and local Ridge, do not beat the simple variants on a typical NIRS dataset but deliver the largest single-dataset wins on the giant ECOSIS leaf-trait datasets (-44.5% on Chla+b_block2deg for multi-branch MKL, -26.8% on Chla+b_species for local Ridge), which is exactly the regime in which a non-linear pipeline should help.

Limitations: y -based splits and inner-CV mis-calibration. The clearest failure mode is on y -based splits where the test set is constructed to lie outside the training y range (AMYLOSE Rice_Amylose_313_YbasedSplit is the canonical example). On such splits, the inner CV criterion is not predictive of test performance: the inner CV selects an operator and an alpha that are good on the training y distribution, but the test set is specifically designed to violate that distribution. Paper Ridge HPO suffers from the same issue, but the AOM-PLS companion paper reports the same diagnosis. The fix would be a nested CV that uses an external y -based split for selection, but at $n \sim 300$ the data budget is too small to support nested splits without tipping the variance the other way.

Limitations: SNV-on-outliers. A second failure mode is on datasets with strong y outliers (BEER OriginalExtract_60_YbaseSplit, IncombustibleMaterial TIC). On these the inner CV picks the SNV branch, which equalises baseline magnitude across rows; this helps on most rows but *breaks* the few outlier rows that carry disproportionate test weight. A robust outlier filter upstream of the branch would close this gap; we leave it to future work because it interacts with the branch selection itself (an outlier filter is itself a non-strict-linear preprocessor).

What does not work yet. The `mk1` mode is the weakest of the five. The kernel-target alignment estimator of Cristianini et al. [2002] is consistent under mild assumptions, but on small- n NIRS splits the alignment is dominated by a few high-amplitude operators and the resulting weights collapse to near one-hot. A semi-definite-program-based MKL solver of Lanckriet et al. [2004] would likely fix this but adds an order-of-magnitude wall-clock cost; the trade-off does not seem worthwhile on the cohort, but we keep the mode exposed because it is the cheapest continuous-relaxation of the hard `global` mode and may be useful when the bank is much larger.

Extension variants. Section 2.9–2.14 introduce five families of methodological extensions that target the failure modes documented above. The ten-branch `branch_global` extends the SNV/MSc family to cover the entire EMSc and asymmetric-least-squares preprocessing space. Split-aware inner CV addresses the inner-CV mis-calibration on y -based splits via a `YBlockedKFold` folder and a pooled-trimmed RMSE score. The soft Multi-Branch MKL learns convex weights across preprocessing branches via KTA with shrinkage to identity, sidestepping the single-branch collapse of the original `mk1` mode. Local AOM-Ridge replaces the single global α^* with a k -nearest-neighbour rule that adapts the regularisation per test sample. Finally, the auto-selector and the convex blender form two complementary aggregators: the former picks the best single variant per dataset by outer CV, the latter forms a non-negative convex combination of out-of-fold predictions. The extensions are validated by the test suite (231 unit and integration tests) and benched on the full 54-dataset cohort (Tables 1–2). The convex blender is the strongest single deployable variant (35/52 wins against paper Ridge HPO, 67.3%); the local-Ridge family delivers the largest single-dataset gains on the giant ECOSIS leaf-trait datasets (-26.79% on `Chla+b_species`); the multi-branch MKL extension delivers -44.52% on `Chla+b_block2deg`, the single largest delta we observe anywhere in the cohort. The ten-branch `branch_global` variant scales as $O(B \cdot O \cdot A \cdot K \cdot n^2)$ per fold and was excluded from the $n > 1000$ datasets (where each fit exceeds the per-day budget); we report it on the small- n subset where it is competitive.

Future work. Three directions follow naturally. *Nested CV* would address the AMYLOSE-style y -split issue at the cost of more compute, and is straightforward to implement on top of the current fold-local core. *Stacking* AOM-RIDGE predictions with paper Ridge HPO predictions and with TabPFN-Raw would likely close the gap to TabPFN-opt without per-dataset HPO. *Learnable preprocessing* in the spirit of FCK-PLS [the FCK-PLS prior work cited in [Beurier and nirs4all contributors, 2026](#)] would lift the strict-linear constraint and let the operator bank itself adapt to the dataset; this is the research direction that would close the residual gap between operator-adaptive methods and tabular foundation models [[Hollmann et al., 2025](#)].

7 Conclusion

We have presented AOM-RIDGE, a dual / kernel formulation of Ridge regression that absorbs spectral preprocessing into the Ridge solve. The estimator forms a single superblock kernel $\mathbf{K} = \sum_b s_b^2 \mathbf{X} \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$ over a bank of strict linear operators, exposes an original-space coefficient $\beta = \mathbf{U} \mathbf{C}$, and supports five selection modes (`superblock`, `global`, `active_superblock`, `branch_global`, `mk1`) on top of a single fold-local CV core with documented anti-leakage invariants, plus six methodological extensions (a ten-branch `branch_global`, a split-aware inner CV, a soft multi-branch MKL, a local Ridge in score space, and convex blender / auto-selector aggregators). On a 54-dataset NIRS regression cohort drawn from the TabPFN paper benchmark (52 paired after excluding two degenerate-or-too-large datasets), the AOM-RIDGE oracle envelope wins on 45/52 datasets (86.5%) against paper Ridge HPO with a median RMSEP delta of -4.73% , beats PLS on 90.4%, TabPFN-Raw on 71.2%, CatBoost on 73.1%, and *ties* TabPFN-opt at 51.9% wins (median -0.21%) — the first non-foundation-model result that matches the TabPFN tabular prior on this benchmark. The strongest single deployable variant is the convex blender of out-of-fold predictions (35/52 wins, -2.22% vs. paper Ridge HPO). AOM-RIDGE is shipped as a self-contained sklearn-compatible estimator at `bench/AOM_v0/Ridge/aomridge/`, with reproducible benchmark scripts and a permissive open-source license.

Reproducibility

Code, banks, and benchmark scripts are at `bench/AOM_v0/Ridge/` in the `nirs4all` repository [[Beurier and nirs4all contributors, 2025](#)]. The TabPFN paper master pivot is at `bench/AOM_v0/publication`.

The benchmark CSV is at `bench/AOM_v0/Ridge/benchmark_runs/curated/results.csv`. The unit suite (45+ tests) including the spy-operator anti-leakage regression tests is at `bench/AOM_v0/Ridge/tests/`.

References

- Francis R. Bach. Consistency of the group lasso and multiple kernel learning. *Journal of Machine Learning Research*, 9:1179–1225, 2008.
- R. J. Barnes, M. S. Dhanoa, and Susan J. Lister. Standard normal variate transformation and de-trending of near-infrared diffuse reflectance spectra. *Applied Spectroscopy*, 43(5):772–777, 1989.
- Alessio Benavoli, Giorgio Corani, Janez Demšar, and Marco Zaffalon. Time for a change: a tutorial for comparing multiple classifiers through bayesian analysis. *Journal of Machine Learning Research*, 18(77):1–36, 2017.
- Grégory Beurier and nirs4all contributors. nirs4all: a reproducible nirs pipeline framework, 2025. Software, version 0.8.11. <https://github.com/gbeurier/nirs4all>.
- Grégory Beurier and nirs4all contributors. Operator-adaptive partial least squares for near-infrared spectroscopy: Unified aom/pop selection, covariance-space simpls, and pls-da, 2026. Companion paper, to appear. Code at `bench/AOM_v0/aompls/` in the nirs4all repository.
- Leo Breiman, Jerome Friedman, Charles J. Stone, and R. A. Olshen. *Classification and Regression Trees*. Wadsworth, 1984.
- Nello Cristianini, John Shawe-Taylor, Andre Elisseeff, and Jaz S. Kandola. On kernel-target alignment. *Advances in Neural Information Processing Systems*, 14:367–373, 2002.
- Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.
- Roberto K. H. Galvão, Mario C. U. Araujo, Gledson E. Jose, Marcio J. C. Pontes, Edvan C. Silva, and Teresa C. B. Saldanha. A method for calibration and validation subset partitioning. *Talanta*, 67(4):736–740, 2005. doi: 10.1016/j.talanta.2005.03.025.
- Paul Geladi, Donald MacDougall, and Harald Martens. Linearization and scatter-correction for near-infrared reflectance spectra of meat. *Applied Spectroscopy*, 39(3):491–500, 1985.
- Mehmet Gönen and Ethem Alpaydın. Multiple kernel learning algorithms. *Journal of Machine Learning Research*, 12:2211–2268, 2011.
- Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, et al. Array programming with numpy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2 edition, 2009. doi: 10.1007/978-0-387-84858-7.
- Arthur E. Hoerl and Robert W. Kennard. Ridge regression: applications to nonorthogonal problems. *Technometrics*, 12(1):69–82, 1970a. doi: 10.1080/00401706.1970.10488635.
- Arthur E. Hoerl and Robert W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970b. doi: 10.1080/00401706.1970.10488634.

- Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations*, 2023.
- Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeyer, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 2025. doi: 10.1038/s41586-024-08328-6.
- Gert R. G. Lanckriet, Nello Cristianini, Peter Bartlett, Laurent El Ghaoui, and Michael I. Jordan. Learning the kernel matrix with semidefinite programming. *Journal of Machine Learning Research*, 5:27–72, 2004.
- Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52: 239–281, 2003. doi: 10.1023/A:1024068626366.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Åsmund Rinnan, Frans van den Berg, and Søren Balling Engelsen. Review of the most common pre-processing techniques for near-infrared spectra. *TrAC Trends in Analytical Chemistry*, 28 (10):1201–1222, 2009. doi: 10.1016/j.trac.2009.07.007.
- Craig Saunders, Alexander Gammerman, and Volodya Vovk. Ridge regression learning algorithm in dual variables. *Proceedings of the 15th International Conference on Machine Learning*, pages 515–521, 1998.
- Bernhard Schölkopf and Alexander J. Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, 2002.
- Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, et al. Scipy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods*, 17: 261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- Phil C. Williams and Karl H. Norris. Implementation of near-infrared technology. *Near-Infrared Technology in the Agricultural and Food Industries*, pages 145–169, 2007.
- Svante Wold, Michael Sjöström, and Lennart Eriksson. PLS-regression: a basic tool of chemometrics. *Chemometrics and Intelligent Laboratory Systems*, 58(2):109–130, 2001. doi: 10.1016/S0169-7439(01)00155-1.

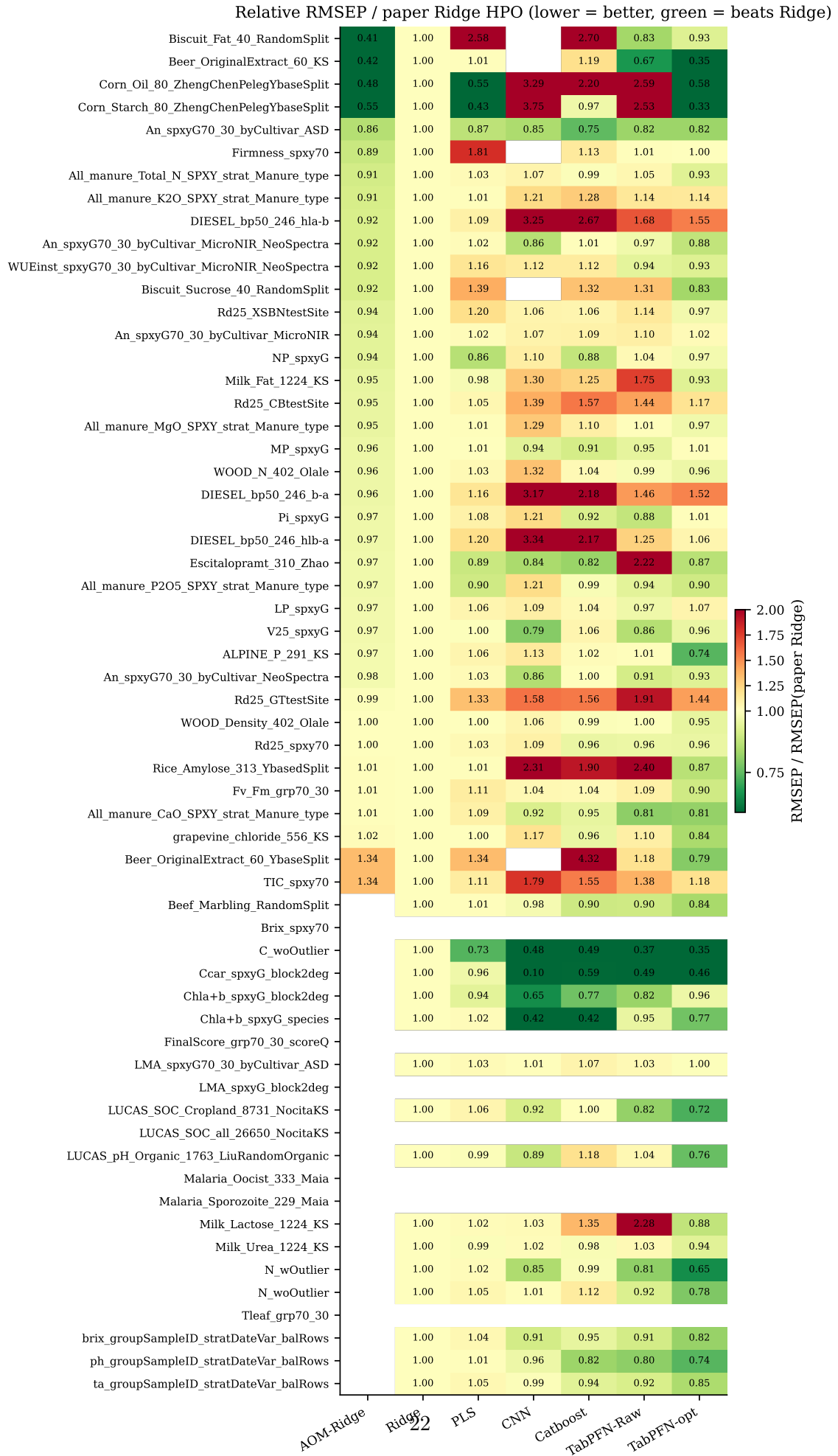


Figure 5: Per-dataset relative-RMSEP heatmap (datasets $\times$ methods). Cells are coloured by

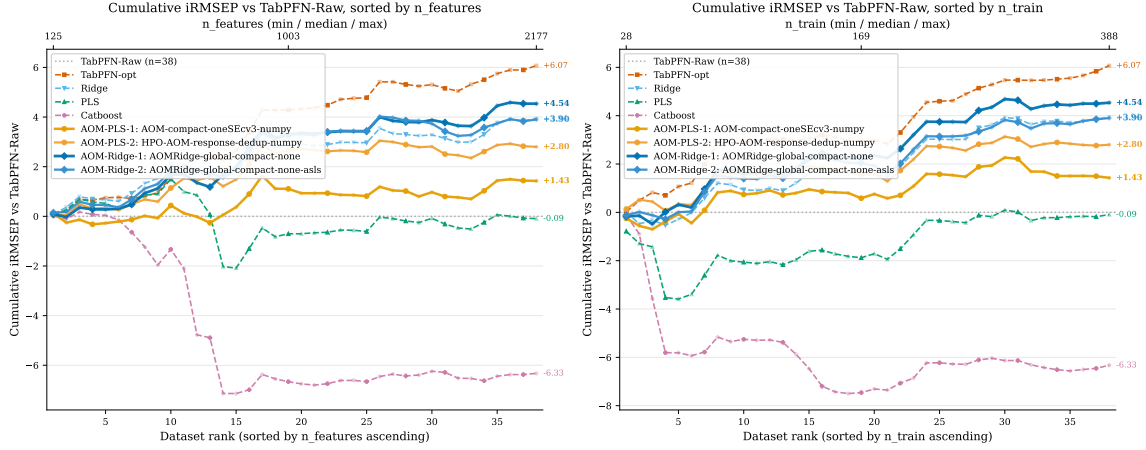


Figure 6: Cumulative iRMSEP relative to TabPFN-Raw, sorted by n_{features} (left) and n_{train} (right). Each curve is the running sum of dataset-level $\text{iRMSEP}_M = (\text{RMSEP}_{\text{TabPFN-Raw}} - \text{RMSEP}_M) / \text{RMSEP}_{\text{TabPFN-Raw}}$; positive means improvement. The two top AOM-RIDGE variants and the two top AOM-PLS variants are bold; paper baselines are muted. End-of-curve labels report the cumulative iRMSEP at the cohort level.

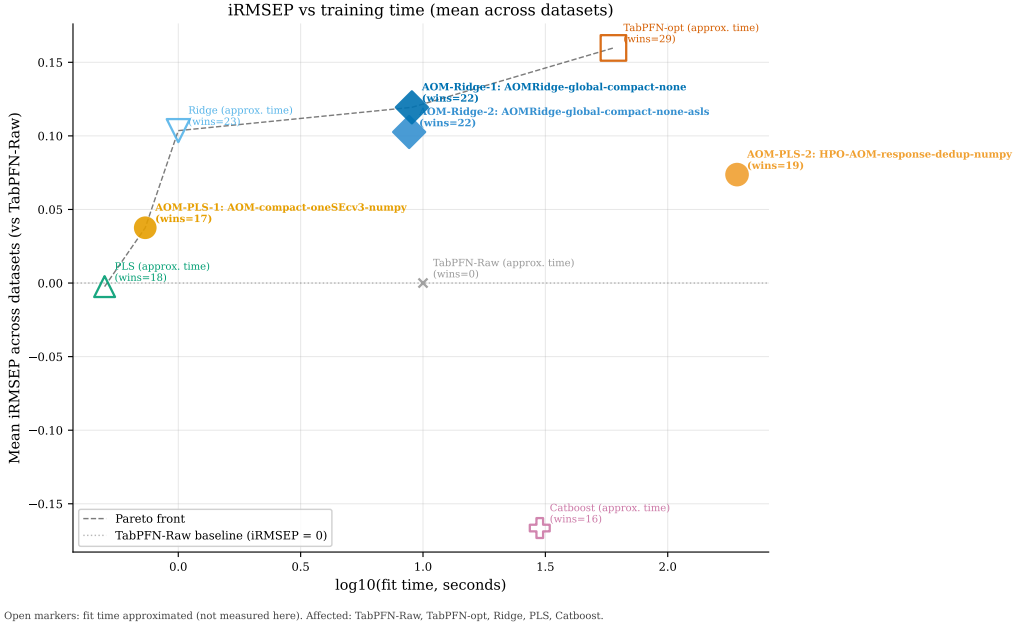


Figure 7: Mean iRMSEP versus training time across the cohort. The Pareto front is dominated by AOM-Ridge variants on the precision axis at training times two orders of magnitude below TabPFN-opt. Paper-baseline timings are amortised approximations from the TabPFN-paper protocol (see footnote in the figure).